

Namit Seth

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Education

Keshav Mahavidyalaya, Delhi University – B.Sc. (H) Electronics

Aug 2023 - June 2027

Technical Skills & Concepts

- **Embedded & Hardware:** Microcontrollers (ESP8266, ESP32, ATmega), RTC (DS1302), Sensor Interfacing (MPU6050), PCB Design (KiCad), Simulation (Proteus).
- **Protocols:** I2C, SPI, UART, NEC (IR).
- **Languages & Software:** C/C++ (Embedded), Python (Automation/Regex), JavaScript, SQL, Git.

Projects

ATmega32 Development Board (PCB Design / KiCad) <https://github.com/namit-seth/ATmega32-Development-Board>

- Designed and fabricated a custom ATmega32 development board, from schematic capture to PCB layout.
- Implemented ICSP interfaces for direct microcontroller flashing and designed an op-amp based 2.5V reference circuit driving a bi-directional LED array to visually debug I/O pin states.

Universal IR Hub (ESP8266 / C++)

<https://github.com/namit-seth/Universal-IR-hub>

- Reverse-engineered undocumented NEC protocol (0xF7 series) payloads to develop a universal control system, replacing generic IR libraries.
- Developed custom C++ firmware to process protocol payloads with optimized memory usage, integrating the hardware into the Google Home ecosystem via Sinric Pro.

DS1302 Embedded Library (C++)

<https://github.com/namit-seth/DS1302>

- Replaced high-overhead standard libraries with a lightweight custom RTC driver authored from scratch using hardware datasheet analysis.
- Implemented bit-banged serial communication and direct register-level control, ensuring stable deployment with a minimal footprint in resource-constrained systems.

Standalone Digital Polling System (Microcontroller)

<https://github.com/namit-seth/EVM-V2>

- Resolved signal bouncing and ghost inputs by engineering software-based debouncing logic, eliminating the need for additional physical hardware capacitors.
- Programmed input-handling logic to detect and reject simultaneous multi-button presses, routing validated totals to a 16x2 LCD interface.

Live Multiplayer Treasure Hunt Game (Flask / SQL)

<https://github.com/namit-seth/treasure-hunt>

- Engineered and deployed a real-time web-based treasure hunt for 10 concurrent teams during a departmental festival.
- Diagnosed critical database locks during simultaneous logins, restoring system stability by restructuring transaction management to resolve race conditions.

Experience

Core Systems Intern, SpaceBorn

Jun 2026 – Present

- Developed foundational embedded firmware skills on **STM32** using STM32CubeIDE/CubeMX, implementing and testing **I2C/SPI** communication wrappers, sensor data structures, UART debugging, and Git-based development workflows.
- Worked on IMU sensor integration fundamentals by studying sensor register maps, data acquisition workflows, and sensor fusion concepts, while validating embedded code through simulation environments such as Wokwi before hardware deployment.

Extracurricular Activities

Secretary, Elektronika, Keshav Mahavidyalaya

2025 – 2026

- Direct the core team in organizing Elexonia, the annual electronics festival, managing technical event logistics and cross-functional student teams.

Executive Member, Student Council, Keshav Mahavidyalaya

2025 – 2026

- Fulfilled administrative duties as an appointed representative, acting as an official signatory to validate student council resolutions.